

EDUCATION

Engineering Degree in Computer Systems 2024 – Present

MedTech University — Tunis, Tunisia

- Specialization: Embedded Systems, Networks, IoT, Robotics & Low-Level Programming
- Strong focus on innovative projects and systems architecture

Integrated Preparatory Classes (MPI) 2022 – 2024

MedTech University — Tunis, Tunisia

- Intensive training in Mathematics, Physics, Computer Science & Engineering Sciences

TECHNICAL SKILLS

Robotics & Simulation	ROS 2 (nodes, SLAM, trajectory planning), LiDAR perception, Gazebo, Rviz 2, autonomous navigation
Embedded & HW	STM32, ESP32, Raspberry Pi 4, Arduino, FPGA (Verilog), PCB Design (Altium)
Protocols	CAN bus, I2C, SPI, UART, Modbus RTU/TCP, Wi-Fi, Bluetooth, MQTT, LoRa, GSM, ESP-NOW, OBD-II
Software	ROS 2, MATLAB/Simulink, Fusion 360, Oracle APEX, Git
Languages (Programming)	C / C++, Python, Java, Verilog
Domains	Embedded Systems, Autonomous Robotics, IoT, Automotive Diagnostics, Embedded AI
Spoken Languages	English (fluent) · French (fluent) · Arabic (native)

PROJECTS & EXPERIENCE

PillPal — Founder & Hardware / 3D Design Lead 2026 – Present

Startup | ESP32, Mobile App, AI, 3D Design — 🏆 1st Place, Best MedTech Project 2024

- Founded PillPal, a startup building a smart medication dispenser, evolving from an award-winning MedTech capstone project (1st place, 2024) into a standalone company
- Built the device around an ESP32, paired with a companion app that notifies users and integrates AI to detect whether a pill has actually been taken
- Automated dispensing for morning, midday, and night doses across a full week, fully configurable through the app
- Responsible for hardware architecture and 3D design of the device enclosure and dispensing mechanism

MiniMakers — Founder 2025 – Present

Robotics School for Kids & Teens (ages 9–15) — Tunis, Tunisia | Arduino, Robotics Education

- Founded MiniMakers, a robotics school teaching kids and teenagers aged 9 to 15
- Taught 30+ students during a summer program covering Arduino programming and hands-on robotics
- Designed and led a flagship project: a four-wheel autonomous car with obstacle detection

Multi-Protocol IoT Gateway — Summer Internship, Tekkim June – July 2026

Embedded Systems Internship — Tunis, Tunisia | ESP32, Modbus RTU/TCP, Wi-Fi, Bluetooth, MQTT, GSM, LoRa, ESP-NOW, Python

- Designed and developed a multi-protocol gateway on ESP32 bridging Modbus RTU/TCP, Wi-Fi, Bluetooth, MQTT, GSM, LoRa, and ESP-NOW communication protocols
- Built an embedded web interface hosted directly on the ESP32 for on-device configuration of gateway and protocol settings
- Developed a Python application to simulate and monitor Modbus registers, enabling end-to-end testing and validation of the gateway without physical field devices

Automotive Diagnostic System — Bako Motors 2025

Technical Project | CAN Protocol, Python, Cloud Dashboard

- Designed and developed an embedded vehicle diagnostic system based on the CAN bus protocol
- Implemented a cloud-connected Python interface for real-time OBD-II monitoring and diagnostics
- Integrated alerts and visualizations to support fault detection and preventive maintenance

Autonomous Vehicle — Research Internship, MedTech SmartLab 2024

Research Internship | ROS 2, LiDAR, Raspberry Pi 4

- Developed a complete autonomous vehicle under ROS 2 with LiDAR perception and navigation in a controlled environment
- Mastered the ROS 2 architecture: inter-node communication, sensor management, SLAM, and trajectory planning
- Validated the system on a real course with an obstacle detection rate > 95%

- Developed an interactive interface for managing and digitalizing banking declaration forms
- Deployed to production for internal use by the Central Bank of Tunisia

CERTIFICATIONS & DISTINCTIONS

Certifications

- PCB Design — Altium Designer (certified)
- Robot Operating System 2 (ROS 2)

Distinctions

- Ranked 1st in Class — 3rd & 4th Year (Computer System Engineering)
- 🏆 Best MedTech Year-End Project 2026
- 🏆 Best MedTech Year-End Project 2025
- 🏆 Best MedTech Year-End Project 2024
- 🥈 2nd Place — TechRevive 2.0 2025
- 🥉 3rd Place — SMU AI Hackathon 2025

LEADERSHIP & AFFILIATIONS

- IEEE-HKN (Eta Kappa Nu) Honor Society Member, Nu Tau Chapter, SMU — since January 2026
- IEEE SMU Student Branch — Treasurer, 2024
- IEEE SMU Robotics & Automation Society (RAS) Treasurer, 2023